SOKOINE UNIVERSITY OF AGRICULTURE

School of Engineering and Technology | Department of Agricultural Engineering

AEN 214: INTRODUCTION TO ELECTRONICS

OPEN COURSEWORK ASSIGNMENT 3: NUMBER SYSTEMS, DIGITAL ELECTRONICS & BOOLEAN KARNAUGH MAPS

Curriculum Coverage: Topic 7 (Number Systems), Topic 8 (Digital Electronics), Topic 9 (Boolean Algebra & K-Maps)
Due Date / Submission: Week 13 of Semester | Individual Submission | Total Marks: 100 Marks

Instructions to Candidates:
1. This assignment is an open engineering coursework assessment designed to develop analytical and practical circuit design competence.
2. Answer ALL twelve (12) questions. Questions are tiered into Part A (Foundational Concepts), Part B (Quantitative Circuit Analysis), and Part C (Agricultural Engineering Systems Design).
3. Show all mathematical derivations, state all engineering assumptions clearly, sketch labeled schematics where applicable, and cite proper SI units.
4. Neatness, logical presentation, and clear engineering justification contribute to technical mastery and exam preparation.

PART A: Radix Conversions & Signed Digital Arithmetic [20 Marks]

Question 1 (5 Marks): Convert the decimal value 219.6875 into: (a) Pure Binary (up to 4 fractional bits), (b) Hexadecimal, and (c) Octal. Show all successive division and multiplication steps clearly.

Question 2 (5 Marks): Perform the following signed 8-bit two's complement additions and subtractions. In each case, state whether an arithmetic overflow occurs and verify by checking the sign-bit carry condition $V = C_{in} \oplus C_{out}$:
(a) (+54) + (+82)
(b) (+75) - (+98) [i.e., (+75) + (-98)]

Question 3 (5 Marks): Perform the BCD addition of decimal numbers 678 + 485 using 8421 BCD arithmetic. Show the binary addition for each decimal digit decade and illustrate where the +6 (0110_2) correction factor must be applied.

Question 4 (5 Marks): An autonomous tractor front-wheel steering shaft encoder outputs angular position in 4-bit Gray code. (a) Convert the Gray code words 1101 and 1010 into standard binary. (b) Explain why Gray code eliminates mechanical and optical reading race hazards relative to standard binary encoding.

PART B: Logic Gate Synthesis & Arithmetic Circuits [35 Marks]

Question 5 (8 Marks): Construct the truth tables and write the Boolean algebraic expressions for 3-input NAND, NOR, and XOR gates. Explain how the XOR gate functions as an odd parity generator for agricultural serial telemetry.

Question 6 (9 Marks): Prove that the NAND gate is a universal logic gate by drawing the exact schematic connections to synthesize: (a) Inverter (NOT), (b) 2-input AND gate, (c) 2-input OR gate, and (d) 2-input NOR gate. Use only standard 2-input NAND gates.

Question 7 (9 Marks): A 1-bit binary Full Adder circuit accepts inputs A, B, and C_{in}. (a) Construct the complete 8-row truth table for outputs Sum (S) and Carry-out (C_{out}). (b) Write the minimized Boolean expressions for S and C_{out}. (c) Draw the logic diagram using two Half Adders and an OR gate.

Question 8 (9 Marks): Using Boolean algebra axioms and De Morgan's theorems, algebraically prove the following identities:
(a) $A + (A \cdot B) = A + B$
(b) $(A + B) \cdot (A + C) = A + B \cdot C$
(c) $(A \cdot B \cdot C) = A + B + C$

PART C: K-Map Minimization & Agricultural Safety Synthesis [45 Marks]

Question 9 (11 Marks) - Automated Silo Fumigation Exhaust Logic:
A grain silo fumigation system has four safety sensors: P (Phosphine Gas Detected), T (Temperature > 35°C), D (Silo Access Door Closed), and E (Emergency Switch Active). The exhaust ventilation blower (B) must turn ON (B = 1) according to the truth function: $B(P, T, D, E) = \sum m(2, 3, 6, 7, 10, 11, 14, 15)$.
(a) Plot this function on a 4-variable Karnaugh Map (Gray code ordered).
(b) Identify all prime implicants and essential prime implicants.
(c) Derive the minimized Sum of Products (SOP) expression.
(d) Implement the minimized expression using NAND-NAND logic gates.

Question 10 (11 Marks) - Grain Dryer Burner Safety Interlock with Don't Cares:
An automated cross-flow grain dryer burner shutoff circuit F(A, B, C, D) is monitored by four binary inputs: A = High Burner Temp; B = Flame Detected; C = Airflow Blower Running; D = Master Control Armed. The shutdown alarm triggers for minterms: $F = \sum m(0, 1, 4, 5, 8, 9, 12)$. Due to mechanical sensor interlocks, input states $d = \sum d(10, 11, 13, 14, 15)$ can never physically occur.
(a) Map the minterms and Don't Care conditions onto a 4-variable K-map.
(b) Form the largest possible groups by leveraging the Don't Care terms.
(c) Write the minimal SOP expression and compare the gate count against a design that treats all Don't Cares as zeros.

Question 11 (11 Marks) - Autonomous Combine Harvester Header Height Controller:
An autonomous wheat combine harvester uses three ultrasonic height sensors located on the left (L), center (C), and right (R) of the cutting header. Each sensor outputs '1' if the cutter bar is too close to the ground (< 10 cm). A hydraulic lift actuator must raise the header (Y = 1) if at least two out of the three sensors indicate ground proximity, OR if an obstacle proximity sensor (O = 1) triggers.
(a) Formulate the truth table for the 4-variable function Y(O, L, C, R).
(b) Plot the function on a 4-variable K-map.
(c) Obtain the minimal SOP expression for hydraulic lift actuation.
(d) Identify any static-1 hazards and add redundant prime implicants if necessary to guarantee glitch-free hydraulic valve switching.

Question 12 (12 Marks) - Digital Agricultural Dosing Pump Controller:
A dairy cattle automated parlor dosing system dispenses liquid nutritional supplement based on 3 binary inputs: A (Animal ID Validated), W (Animal Weight > 450 kg), and M (Milk Yield > 25 Liters/day). The dispenser valve V operates according to: $V = A \cdot (W \cdot M + W \cdot M) + A \cdot W \cdot M$.
(a) Simplify the Boolean expression algebraically to its absolute minimum literal count.

(b) Verify your algebraic minimization by constructing a 3-variable Karnaugh Map.
(c) Draw the final simplified circuit using standard logic gates.
(d) If an active-LOW relay module is used to drive the pump, express the required logic function using De Morgan's theorem.